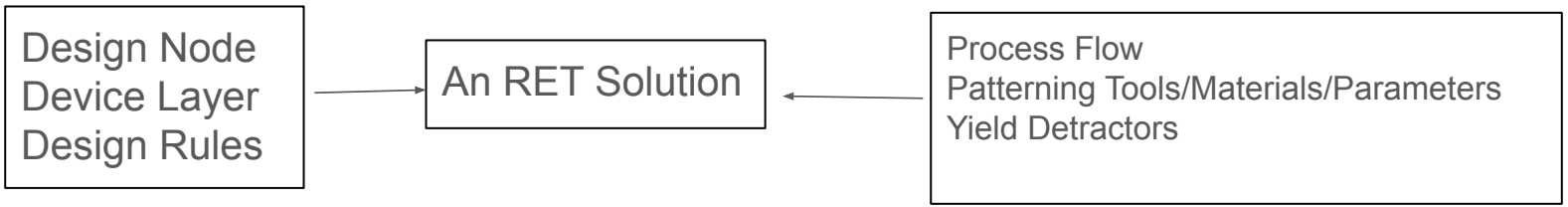
Chapter 1: RET in Practice

From “Photomask Fundamental,” we learned many concepts of RET. Now, let’s have an overview of how RET works in reality.

For a chip manufacturer, i.e., a fab, to qualify for mass production of certain technology node, say 14nm, yield should be above 90% within the constraint of cost and time. After qualification, yield must be constantly monitored and continuously enhanced.

To say that yield is crucially dependent on quality of patterning is an understatement. In fact, process development for patterning of critical layers, i.e., poly, contact, etc., is a make-or-break battle, which cannot afford but to employ the most powerful RET tools.

RET engineers are called upon to deliver RET solutions which enable new design node, new device layer, new process flow, especially new patterning tools/materials/parameter, or to address certain yield detractors, commonly known as hotspots, that have been detected on silicon.



EDA Platform

As we shall see later, an RET solution is mainly computer algorithms which convert a design layout to a mask layout.

Industrial-scale RET algorithms can not be created from scratch. They must be built upon certain EDA platforms. A platform provides modules of libraries for certain computations, such as layout data processing, optic simulations, etc. RET engineers utilize these modules and provide required inputs, for example, design and process information, target of a solution, and options to achieve the solution.

Currently major RET platforms are Proteus by Synopsys, Calibre by Simons (formerly Mentor Graphics), and Tekayan by ASML. RET algorithms are platform specific, that is, they can not be transferred from one platform to another.

It is important to understand that, with same inputs of design/process/target, RET solutions developed on different EDA platforms would deliver similar but never identical masks. A fab can run split lots with masks built on different EDA platform and compare yield. Once the choice is finalized, EDA platform becomes part of process-of-record.

RET Solution Development Cycle

A cycle of developing an RET solution is typically as follows:

Steps	Job	Owner
1	Collect design/process/target information.	RET, Patterning, Design
2	Design test patterns and collect wafer data. As we shall learn later, RET algorithms need to use empirical data.	RET, Patterning
3	Develop RET algorithms. Apply the algorithms on test chips composed of real, production designs as well as test patterns. Order post-RET masks from mask shops.	RET
4	Run silicon with the masks and collect yield data as well as wafer data.	Patterning, Yield analysis
5	Analyze yield data to decide if patterning process and RET solution needs to be improved. This is the end of one cycle.	Yield Analysis, RET, Patterning

Making and Execution of an RET Solution

Most RET engineers work either in a fab or in an EDA company. The former are the customers of the latter. The rest of them work in research institutes and universities.

An EDA platform includes various products. Each is a tool for a specific RET job, for example, layout review, model building, OPC recipe creation, etc. As a fab contracts an EDA company, these tools will be installed on fab computer systems. Once EDA tools are installed, it is the responsibility of fab to develop RET solutions and integrate them into the flow of tapeout. However, making and execution of RET solutions is often a collaboration between the two.

RET engineers in the fab need to go through the learning curve of the tools, through trainings provided by EDA company, reading documents, and, most importantly, practice. AI is also gaining momentum in assisting learning as well as using the tools. They should be able to operate all the tools independently. Picking up EDA tools isn't an easy task. It would take years to become an expert.

Making and Execution of an RET Solution (continued)

EDA companies support their customers directly through field engineer and indirectly through application engineers. The former usually have access to customer system so they can work on customer's secured data. The latter usually owns a particular tool and can be consulted for details that the former can not provide.

It worths pointing out that interactions between customer and EDA support are crucial to the success for both. Behind successful, on-time tapeouts in fab, especially those with adoption of a new RET solution, are EDA's involvement in debugging, temporary software patch-up, testing new flows/functions/features, etc. Behind every new tool and release update of EDA softwares are bug fixing/enhancement requests from customers, real layout cases and fab process conditions provided by customers, and, most importantly, roadmap and strategies for advanced patterning technologies shared between the two parties.

Real example

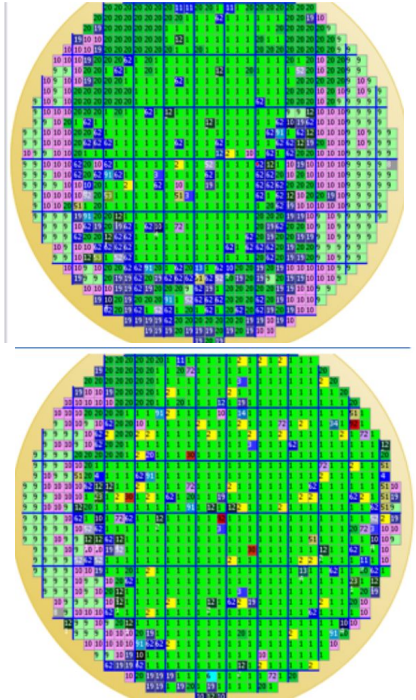
A real example: The author used to work for a 90nm fab. For 1st level metal, contact, and via layers, the fab had been using 249nm stepper, phase shift mask, and model-based OPC built with Calibre, for each layer, respectively. At a certain date, supply of existing photoresist would come to the end due to environmental issues. The fab must switch to a new type of photoresist to meet environmental regulations.

With the new resist and masks built with the old RET solutions, the fab experienced serious yield loss. The author was tasked to develop two RET solutions with a) using 193nm stepper and binary mask, and b) using 249nm stepper and phase shift mask. Eventually, the latter was chosen.

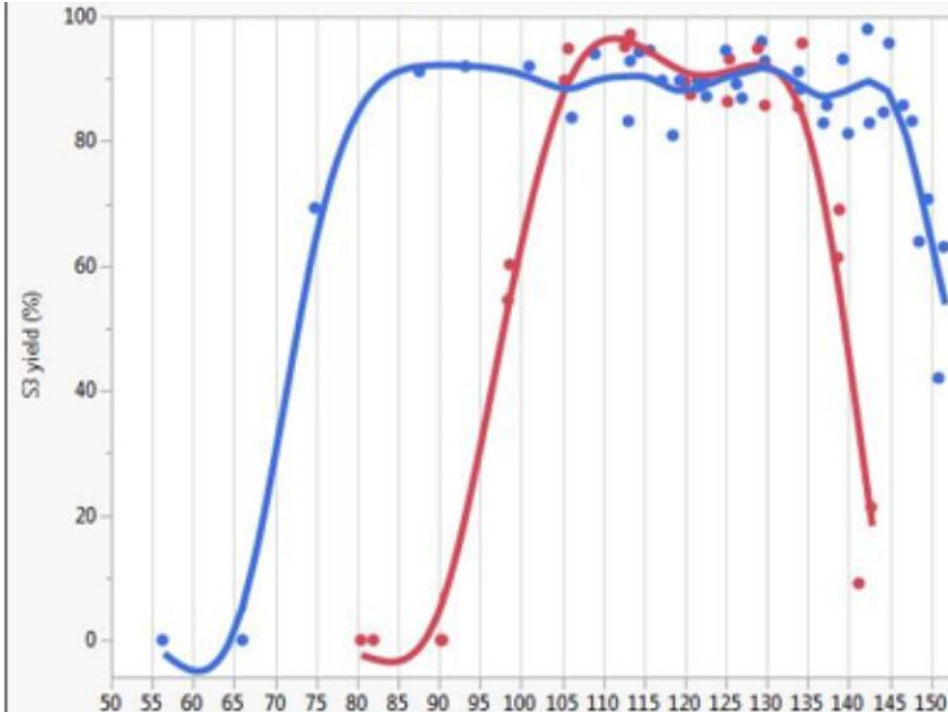
Development of new RET solutions went through two cycles. At the end of the 2nd cycle, the process window for two production chips using the new RET solution doubled that using the old solution (see next slide). The new RET algorithms, including model-based OPC and post-OPC verification, were adopted as Process-of-Record and integrated into the mask data prep flow of the fab.

Example: RET Solution Enhance Yield Significantly

Good dies (light green) on wafer w. old RET solution (up) and w. new RET solution (bottom)



Process Window w old RET solution (red) and w. new RET solution (red)



Summary

In summary, any RET solution is a proprietary property of a fab, which is developed with specific process environment and aims to solve specific yield issues of the fab.

An RET solution is also specific to technology node and device layer. Critical device layers requires more aggressive RET solutions than non-critical ones.

On a production-chip, RET jobs could require powerful, distributed processors by hundreds, and take upto days to complete.

An RET solution is built upon a specific EDA platform. In fact, even with the same platform, an RET solution must use a specific version/release of software. Competitions between EDA companies to win business from fabs are fierce. It is common for some fabs to use different EDA platforms to build RET solutions for different technology nodes or even just different device layers.